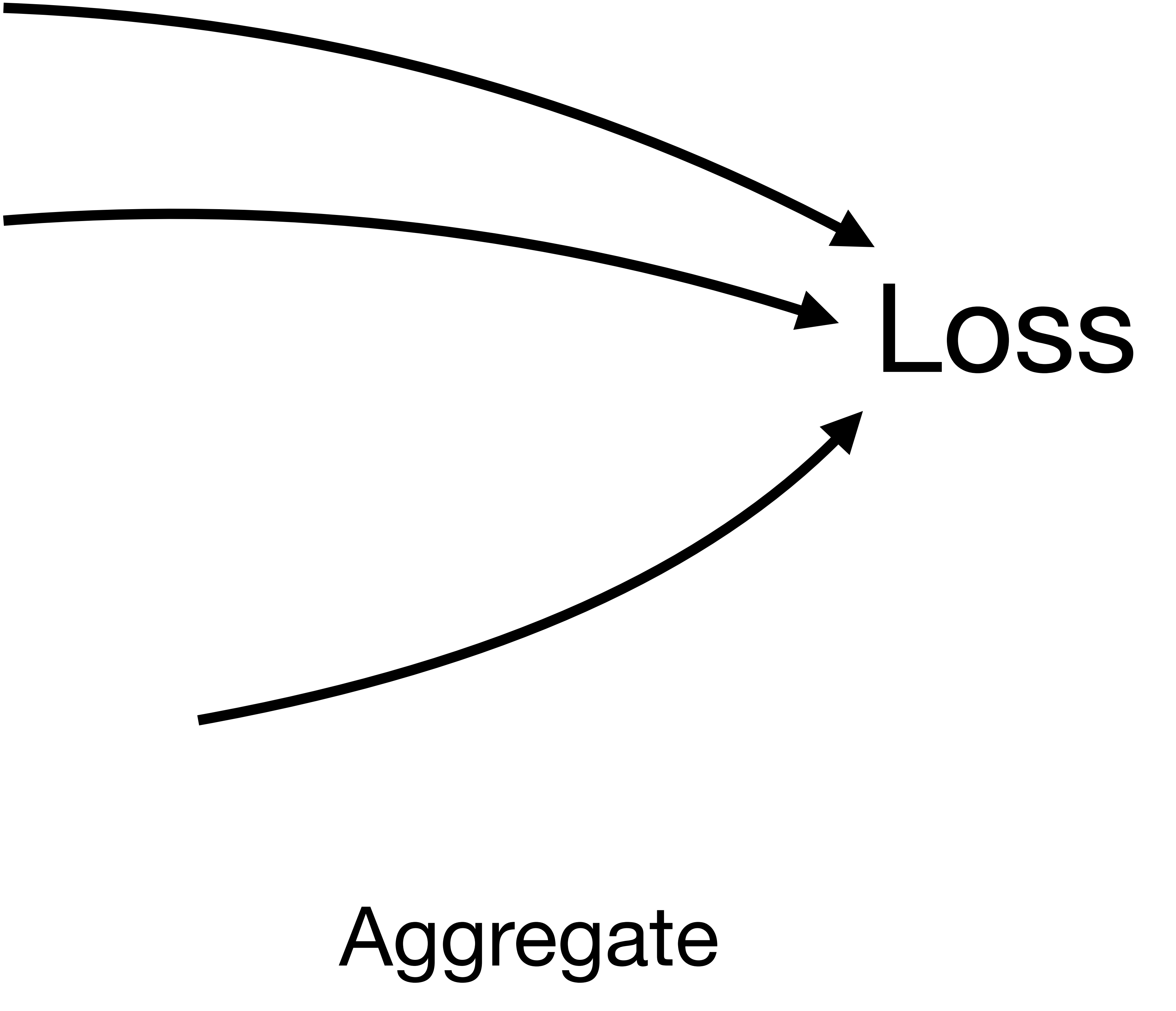
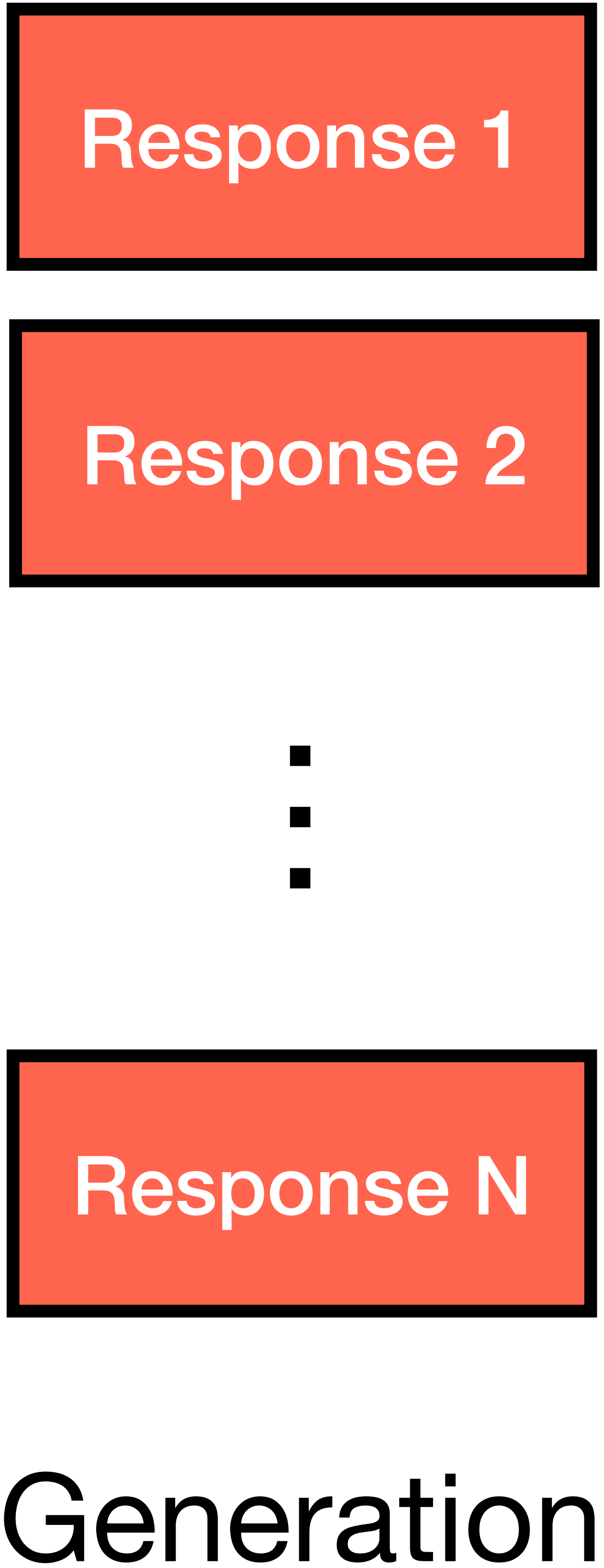
Nate Woodward

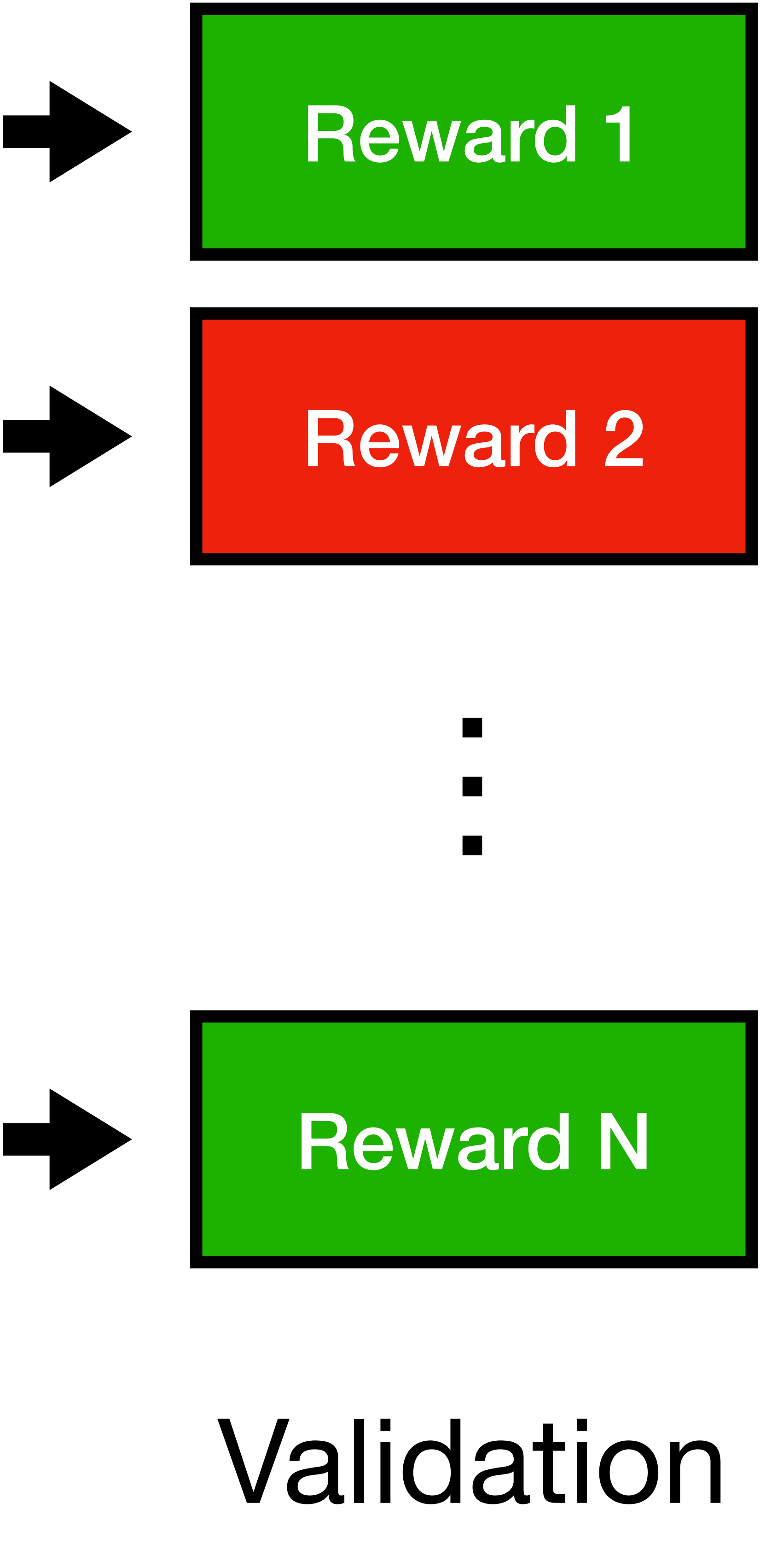
LCTP Spring Symposium: Theoretical Physics and AI


Loss
Aggregate

Reasoning Fine-Tuning: Outcome RL

7

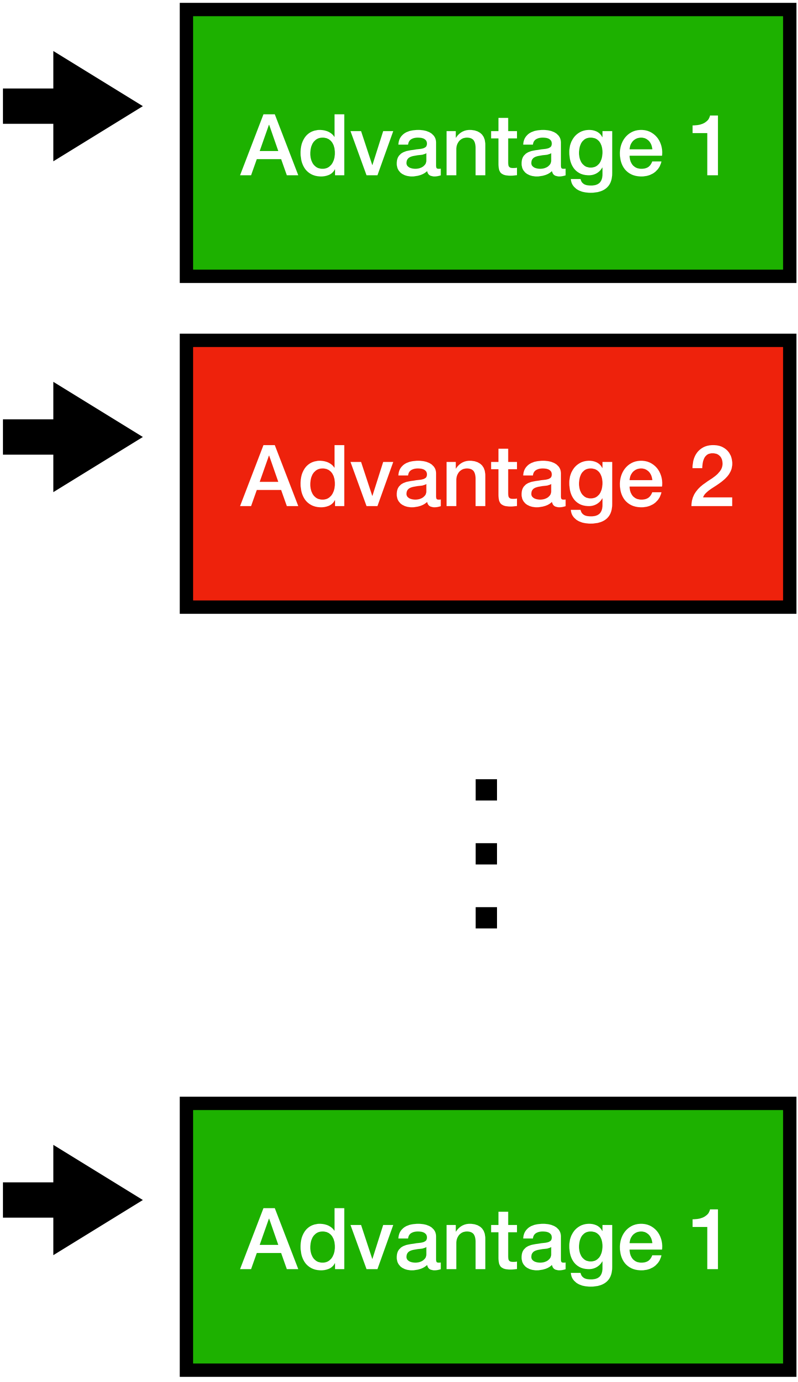

Response 1
Response 2
Response N
Generation

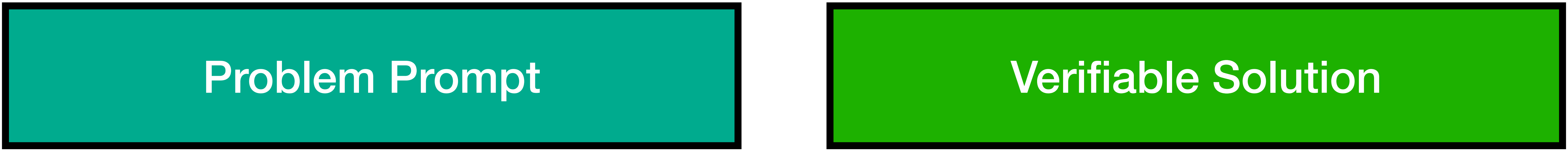

Reward 1
Reward 2
Reward N
Validation

$-\mu$

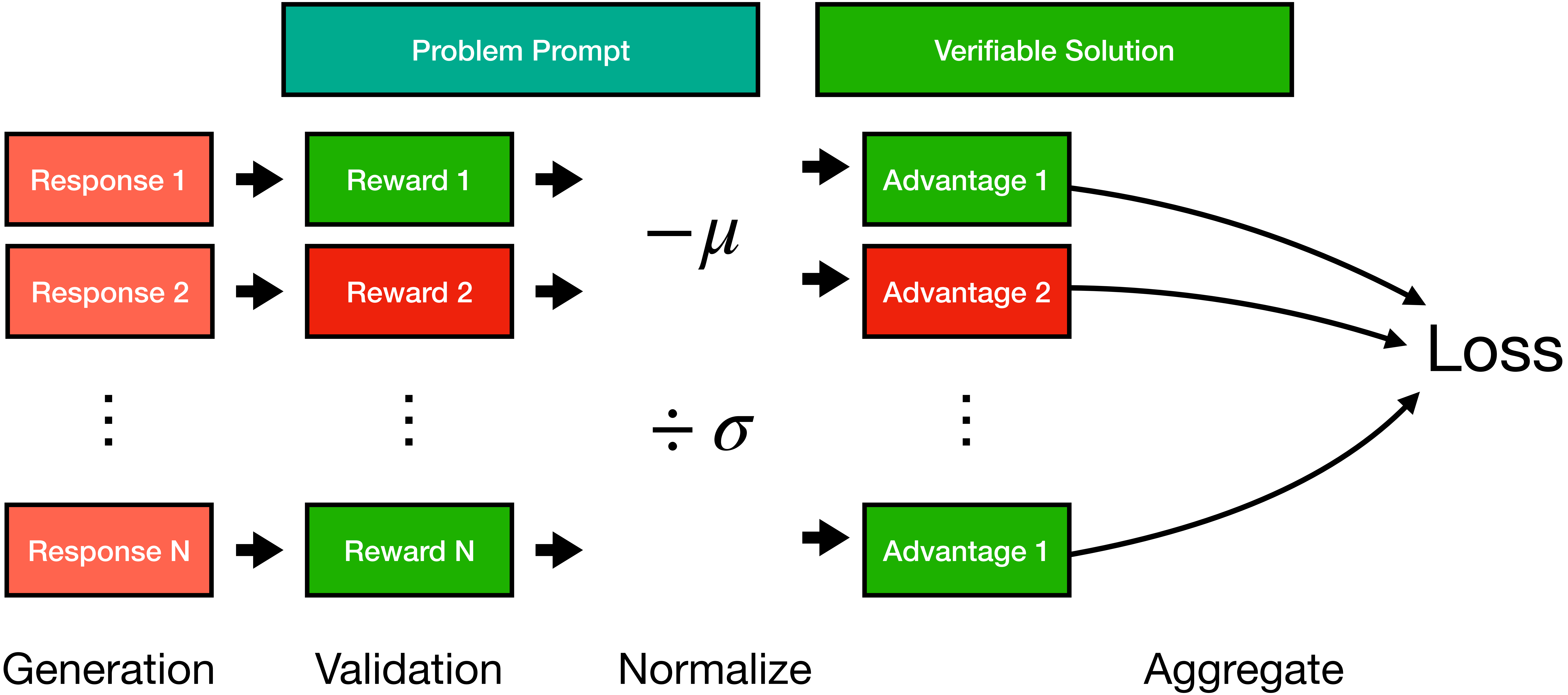
$\div \sigma$

Normalize


Advantage 1
Advantage 2
Advantage 1


Problem Prompt
Verifiable Solution

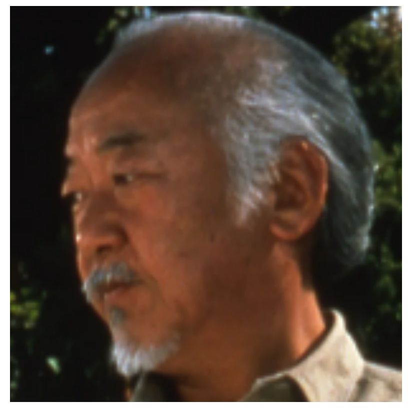
Reasoning Fine-Tuning: Outcome RL



Reasoning Fine-Tuning: SFT

Problem Prompt (X) | Teacher Model Solution

Teacher Solution per token



Y_1 | Y_2 | Y_3 | Y_4 | $\cdots$ | Y_T

Loss

$$-\log P(Y|X) = -\sum_{t=1}^{T} \log P(Y_T|X, Y_{<t})$$

Student Probability per token

$P(Y_1|X)$ | $P(Y_2|X, Y_1)$ | $\cdots$ | $P(Y_T|X, Y_{<T})$